

PTV VISSIM and DhakaSim (Python): A Comparative Assessment

Capabilities, Distinctive Strengths, and a Path to Professional Parity

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Abstract

PTV VISSIM is the microscopic traffic simulator most widely used by government and consulting organisations in Bangladesh. DhakaSim is a research simulator built specifically for the non-lane-based, heterogeneous traffic of cities such as Dhaka, recently reimplemented in Python and extended with real survey data. This report compares the two: what each does well, where they genuinely differ in modelling philosophy rather than merely in polish, and what DhakaSim would need in order to support the professional workflow that VISSIM users expect. The comparison of professional practice follows the workflow set out in Dr. Md. Hadiuzzaman's lecture *Microscopic Simulation Study* (BUET). The conclusion is that the two tools are complementary rather than competing: VISSIM is the stronger engineering platform, while DhakaSim models a traffic regime VISSIM does not represent. The recommended direction is therefore not to imitate VISSIM, but to add the workflow and evaluation machinery that would make DhakaSim's distinctive model usable and credible in professional studies.

1 Scope and Basis of Comparison

This document compares VISSIM against DhakaSim *as currently developed* — the Python reimplementations together with the extensions described in Section 3. Statements about professional simulation practice follow the lecture's workflow: project definition, data collection, network coding, error checking, calibration and validation, scenario design, simulation runs, result analysis, and documentation. No code was modified in preparing this report; it is an assessment, not a change.

2 The Professional Workflow VISSIM Supports

The lecture frames a simulation study as a disciplined engineering process, and VISSIM is built around it:

- **Data preparation:** geometry from construction drawings, aerial photos and grades; controls from the operating agency's signal timings; existing demand as turning volumes, OD tables and vehicle mix; calibration data such as speeds, queues and travel times; and forecast future demand.
- **Network coding:** importing and scaling an aerial or CAD overlay, using coding templates, laying links and nodes over the image, then coding link attributes (lanes, free-flow speeds), intersection attributes (control type and parameters, turn-lane designations, stop bars, turn pockets), source/sink zones, vehicle types and OD tables.
- **Error checking:** verifying connectivity, geometry against the aerial photo, controls, prohibited turns and lane restrictions, zone volumes against counts, and vehicle-mix proportions — assisted by practical strategies such as colour-coding links by attribute, loading 50% demand to expose coding errors, following a single vehicle to spot unrealistic braking or lane changes, and testing simple hand-computable cases.

- **Calibration and validation:** treated as two distinct problems — *supply side* (capacity and saturation flow; global car-following, lane-changing and gap-acceptance parameters; then link-specific fine tuning) and *demand side* (route choice, driver familiarity, link cost functions, free-flow speed) — with explicit calibration targets, validation against the fundamental diagram and field observation, and search algorithms ranging from golden-section for a single parameter to contour plots and genetic algorithms for two or more.
- **Result analysis:** animation output (snapshots, hotspots, dynamic link colouring, vehicle traces) alongside numerical output (volumes, travel time, speed, delay, stops, density, queues), summarised through system indicators such as VMT and VHT, localised indicators such as queue-overflow and hotspot reports, and proper statistics including means, variances and hypothesis testing.
- **Bias correction:** discarding a warm-up period, keeping congestion inside the study area and simulation period, and counting blocked or unreleased vehicles.

3 DhakaSim as Currently Developed

DhakaSim models each road as a set of thin lateral *strips* rather than lanes; a vehicle occupies several strips and may move to any free strip. This is the core design decision, and it is what makes the tool able to represent traffic that has no lane discipline. The present build provides:

- a validated Python reimplementation of the original Java simulator, with a numeric-compatibility layer reproducing Java arithmetic exactly, pinned by a regression test;
- thirteen vehicle types spanning motorised and non-motorised traffic, and side-friction modelling: parked cars, rickshaws and CNGs, standing pedestrians, pedestrians walking along the road, and road-crossing pedestrians;
- a library of thirteen car-following models and four lane-changing models, selectable per run;
- five real intersections built from 24-hour classified turning-count surveys — the Kakrail corridor (two junctions), Bijoy Sarani, Khamarbari, Banani 23 and Banani 27 — each with hourly demand and an hourly vehicle mix, selectable from the GUI along with the hour of day;
- an animated GUI with pan and zoom, per-type vehicle colours, named nodes, a collapsible legend and a live per-type vehicle counter;
- an automatic per-run HTML report containing headline metrics, a configuration table, per-type results, charts, an embedded run animation and a glossary, plus raw CSV outputs.

4 Head-to-Head Comparison

Aspect	PTV VISSIM	DhakaSim (Python)
Traffic regime	Lane-based; heterogeneity supported but within lane discipline	Non-lane-based by construction — strip occupancy, free lateral movement
Vehicle population	User-defined classes, typically motorised	13 types including rickshaw, van/cart, CNG, bicycle; mix measured per site and hour
Side friction	Not represented as a first-class phenomenon	Explicitly modelled: parked vehicles, standing/walking/crossing pedestrians

Aspect	PTV VISSIM	DhakaSim (Python)
Car-following	Wiedemann 74/99 psycho-physical models	13 selectable models , including a strip-based weighted model
Lane changing	Rule-based with gap acceptance	4 selectable models (naive, GHR, Gipps, MOBIL)
Network coding	Graphical editor over scaled aerial/CAD background; templates; turn pockets, stop bars	Hand-edited text files; no editor, no background image
Demand input	OD matrices, turning volumes, dynamic assignment, time-sliced demand	OD demand per hour from survey counts; fixed shortest-path routes, constant rate within a run
Signal control	Fixed-time, actuated, VAP/RBC logic, priority, external controllers	Fixed round-robin phase change only (adaptive control present but disabled)
Calibration support	Documented supply/demand strategy; parameter sets; multi-run seeds	Manual; single legacy corridor calibration; seed handling prevents repeatability
MOEs	Volumes, speed, delay, stops, density, queues, travel time, VMT/VHT, node evaluation	Speed, waiting time, trip time, fuel, flow, per-type and per-route CSVs; no queue, density, delay, VMT/VHT
Statistical rigour	Multi-run replication, warm-up handling, confidence intervals	Single run; no warm-up exclusion, no replication or interval estimates
Visual analysis	2D/3D animation, dynamic link colouring, hotspots, vehicle traces, video export	2D animation, per-type colours, live counter, embedded SVG run animation
Scenario management	Scenario manager, alternatives, comparison reports	One network and hour per run; no scenario comparison
Extensibility	COM API, external driver models, scripting	Open Python source , directly modifiable and scriptable
Cost / access	Commercial licence	Free, no dependencies , standard-library Python

5 What Makes DhakaSim Distinctive

Three of DhakaSim's properties are not matters of maturity, and would not be obtained by adopting VISSIM:

1. **It models a traffic regime VISSIM does not.** VISSIM assumes lane discipline as its organising principle. Where vehicles of very different sizes and speeds share an undivided carriageway and position themselves freely across it, the lane abstraction misrepresents the mechanism of congestion, not merely its magnitude. DhakaSim's strips represent that behaviour directly.
2. **Side friction is a modelled phenomenon, not a fudge factor.** In professional practice the effects of parked vehicles and pedestrian interference are usually absorbed into capacity adjustment factors. DhakaSim generates those elements explicitly, which is what allows their contribution to be

studied rather than assumed.

3. **It is open, inspectable and free.** Every behavioural rule can be read and changed; thirteen car-following models can be swapped and compared; there is no licence barrier for a university or a public agency. This is also what makes it a viable research platform for machine-learning work, which a closed commercial tool is not.

6 Where VISSIM Is Ahead

The gaps are concentrated in engineering workflow rather than in traffic physics: network construction is manual and error-prone without a graphical editor or a scaled background image; signal control is limited to fixed round-robin phasing, which rules out the signal-design studies that motivate many professional projects; the standard intersection performance measures — queue length, delay, density, stops, VMT/VHT — are absent; and there is no support for the statistical hygiene the lecture treats as mandatory, namely warm-up exclusion, multi-run replication and confidence intervals. The last point is aggravated by a known defect: the random seed in the parameter file is ignored, so headless runs are not reproducible.

7 Improvement Roadmap

The following recommendations are ordered by the ratio of professional value to implementation effort. They are intentionally framed so that none of them compromises the model's validated numerical equivalence with the original Java implementation — any change to vehicle movement rules should be made deliberately and documented, not as a side effect of adding features.

Tier 1 — Credibility (highest value, modest effort)

1. **Honour the random seed and add multi-run replication.** Make runs reproducible, then add an n -replication mode that reports the mean and a confidence interval per measure. Without this, no calibration result and no scenario comparison is defensible.
2. **Add warm-up handling.** Exclude an initial loading period from the statistics, as the lecture requires, and report the excluded interval.
3. **Add the missing MOEs:** queue length and queue-overflow reporting, control delay, stops, density, and network totals VMT and VHT. These are the measures agencies expect, and most are derivable from state the simulator already tracks.
4. **Repair the accident and collision counters,** which currently report zero while events are written only to the log.

Tier 2 — Workflow (high value, moderate effort)

5. **Signal control worthy of the name:** per-approach phases with green splits, cycle length and offsets, defined in an input file; then re-enable and validate the existing adaptive controller. This unlocks the signal studies that dominate professional practice, and connects directly to the group's NSGA-II signal-scheduling research.
6. **Scaled background imagery and a network editor.** Even a read-only georeferenced background image behind the animation would let networks be checked against reality; a simple graphical editor for nodes and links would remove the main barrier to building new sites. The four synthetic star geometries currently in use are the clearest symptom of this gap.

7. **Error-checking tools** following the lecture’s checklist: connectivity and demand/route consistency checks, a reduced-demand diagnostic mode, and single-vehicle tracing to expose implausible braking or strip changes.
8. **Time-varying demand within a run.** Demand is currently constant for the whole run, so an hour is simulated as steady conditions. Supporting demand profiles that change during a run would permit peak-spreading studies.

Tier 3 — Analysis and Presentation (valuable, larger effort)

9. **Scenario management:** named scenarios with automatic side-by-side comparison in the HTML report — the natural extension of the existing per-run reporting, and the format agencies actually consume.
10. **Richer visual diagnostics:** dynamic link colouring by speed or density, hotspot highlighting, and vehicle trajectory traces, all of which the lecture lists as standard animation output.
11. **Validation against the fundamental diagram:** flow–density–speed plots from simulated data compared with field observation, which is the lecture’s stated validation criterion and is currently absent.
12. **Automated calibration.** The lecture’s own progression — golden section, contour plots, genetic algorithms — is the conventional route. Given the simulator’s runtime, a surrogate-assisted approach is preferable, and this aligns with the planned MSc work on machine-learning-based calibration.

Deliberately Not Recommended

Pursuing 3D visualisation, a COM-style external API, or full dynamic traffic assignment would consume disproportionate effort for little research return, and would move DhakaSim toward competing with VISSIM on VISSIM’s own terms. The comparative advantage lies elsewhere.

8 Priority Summary

Improvement	Value	Effort	Unlocks
Seed fix + replication + warm-up	High	Low	Defensible results
Queue, delay, density, VMT/VHT	High	Low–Med	Agency-standard reporting
Phase-based signal control	High	Medium	Signal design studies
Background image / network editor	High	Medium–High	Realistic new sites
Error-checking toolkit	Med	Low	Trustworthy coding
Time-varying demand	Med	Medium	Peak spreading
Scenario comparison	Med	Medium	Alternatives analysis
Fundamental-diagram validation	High	Medium	Model credibility
Automated calibration	High	High	Thesis contribution

Table 2: Improvement priorities by value and effort.

9 Conclusion

VISSIM is the more complete engineering platform, and for a conventional lane-disciplined study in Bangladesh it remains the appropriate tool. DhakaSim is not a weaker substitute for it; it is a different instrument, one that represents non-lane-based heterogeneous traffic and its side-friction elements in a way VISSIM structurally cannot, and it is open and free where VISSIM is closed and licensed.

The most productive direction is therefore not to replicate VISSIM's feature list but to close the workflow gap around DhakaSim's distinctive model: make results reproducible and statistically sound, report the performance measures practitioners expect, give it credible signal control, and reduce the cost of building an accurate network. Those changes would let DhakaSim's model be taken seriously in professional studies, which is precisely where the value of a Dhaka-specific simulator lies.

Reference: Md. Hadiuzzaman, *Lecture 4: Microscopic Simulation Study*, Department of Civil Engineering, BUET.